

MONROE, WI, USA

WMO: 726414

Lat: 42.615N

Lon: 89.591W

Elev: 331

StdP: 97.41

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.9	(c) -17.8	(d) -25.0	(e) 0.4	(f) -20.0	(g) -22.5	(h) 0.5	(i) -17.3	(j) 12.7	(k) -2.9	(l) 11.6	(m) -5.8	(n) 4.5	(o) 270	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	32.0	22.7	30.1	22.0	28.7	21.5	24.8	28.8	23.9	27.9	22.9	26.9	5.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.3	26.9	22.5	18.0	26.0	22.0	17.4	25.6	77.3	29.0	73.2	28.0	69.5	27.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.2	(b) 9.5	(c) 8.4	(e) -24.4	(f) 33.6	(g) 4.0	(h) 1.6	(i) -27.2	(j) 34.8	(k) -29.6	(l) 35.8	(m) -31.8	(n) 36.7	(o) -34.7	(p) 37.9		
(4) 11.2	(5) 9.5	(6) 8.4	(7) -24.7	(8) 26.5	(9) 3.9	(10) 0.8	(11) -27.5	(12) 27.1	(13) -29.8	(14) 27.6	(15) -32.0	(16) 28.0	(17) -34.8	(18) 28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-6.1	-4.8	2.2	8.7	15.1	20.5	22.2	21.4	17.7	10.4	3.6	-3.2	(6)
(7)		DBStd	11.32	6.55	6.31	6.45	5.38	4.90	3.41	3.08	2.90	4.25	5.12	5.52	6.06	(7)
(8)		HDD10.0	1942	501	416	256	86	10	0	0	0	2	58	203	411	(8)
(9)		HDD18.3	3777	759	649	502	291	124	17	4	7	62	249	443	669	(9)
(10)		CDD10.0	1593	0	1	13	48	169	314	379	354	232	72	10	1	(10)
(11)		CDD18.3	387	0	0	1	3	26	82	125	103	42	5	0	0	(11)
(12)		CDH23.3	3122	0	0	7	36	226	671	1053	753	333	43	0	0	(12)
(13)		CDH26.7	878	0	0	0	6	58	191	330	201	87	7	0	0	(13)
(14)	Wind	WSAvg	3.9	4.4	4.4	4.3	4.6	4.0	3.3	3.0	2.7	3.1	3.9	4.3	4.2	(14)
(15)	Precipitation	PrecAvg	943	35	39	56	96	100	144	101	110	91	69	59	45	(15)
(16)		PrecMax	1197	85	90	143	170	221	324	208	377	187	145	152	118	(16)
(17)		PrecMin	699	8	2	12	30	15	31	31	36	24	13	4	12	(17)
(18)		PrecStd	126	19	24	37	41	46	70	47	73	41	38	39	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	13.8	23.9	27.5	32.1	32.7	35.2	32.1	27.5	20.0	15.0	(19)	
(20)			MCWB	7.1	9.6	17.1	16.7	20.6	23.0	23.0	22.4	21.8	19.4	14.2	13.2	(20)
(21)		2%	DB	6.4	7.9	17.7	23.0	27.8	31.0	31.7	30.9	28.9	23.9	17.1	9.8	(21)
(22)			MCWB	4.1	5.6	13.5	14.6	18.8	22.0	23.5	22.6	21.2	17.5	12.7	7.6	(22)
(23)		5%	DB	3.7	5.2	14.9	20.9	26.1	28.8	30.0	28.8	27.3	21.4	13.9	7.0	(23)
(24)			MCWB	2.5	3.1	11.2	13.3	17.7	21.1	22.6	21.8	19.9	16.0	11.1	5.8	(24)
(25)		10%	DB	2.3	2.7	11.3	17.7	23.0	27.4	28.0	27.4	25.1	18.6	12.1	4.1	(25)
(26)			MCWB	1.5	1.5	7.8	11.8	16.9	20.3	21.7	20.9	18.6	14.4	9.5	2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	10.1	17.3	18.6	22.5	25.2	26.7	25.6	24.3	21.6	15.8	13.9	(27)
(28)			MCD B	8.5	12.8	22.2	23.9	28.2	30.0	29.9	29.4	28.7	25.6	18.2	15.1	(28)
(29)		2%	WB	4.5	6.0	15.0	16.3	20.6	24.0	25.3	24.4	22.6	19.2	13.6	8.0	(29)
(30)			MCD B	6.3	8.0	17.9	21.2	25.3	28.2	28.5	28.0	26.6	22.2	15.8	8.5	(30)
(31)		5%	WB	2.7	3.5	11.9	14.6	19.4	22.9	24.0	23.3	21.4	17.1	11.9	5.7	(31)
(32)			MCD B	3.8	5.1	14.4	19.8	23.8	27.1	28.3	26.9	24.8	19.6	13.8	6.7	(32)
(33)		10%	WB	1.2	1.8	7.9	12.4	18.1	21.7	22.9	22.3	20.2	14.9	9.1	3.1	(33)
(34)			MCD B	2.1	2.8	10.5	17.0	22.0	25.7	27.2	25.8	23.6	18.4	11.7	3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	8.6	10.5	10.8	10.3	10.1	9.9	10.7	9.6	8.1	6.8	(35)
(36)			MCD BR	8.4	9.6	12.5	14.8	13.0	11.8	11.4	11.5	12.7	12.7	11.4	9.0	(36)
(37)			MCWBR	7.1	7.4	8.0	8.4	6.5	5.9	5.2	5.1	6.0	7.1	8.2	7.5	(37)
(38)		5% WB	MCD BR	8.2	9.3	11.3	12.9	11.2	10.5	10.4	9.7	10.4	11.1	10.6	8.7	(38)
(39)			MCWBR	7.0	7.4	8.4	8.7	6.7	6.2	5.8	5.6	5.9	8.3	8.6	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.301	0.326	0.370	0.397	0.414	0.426	0.405	0.376	0.333	0.294	0.277	(40)	
(41)		taud	2.387	2.334	2.400	2.298	2.276	2.266	2.230	2.290	2.349	2.472	2.514	2.433	(41)	
(42)		Ebn at Noon	867	899	919	896	877	860	844	852	852	855	849	838	(42)	
(43)		Edh at Noon	78	98	104	125	131	133	137	125	109	85	69	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.73	2.73	3.75	4.62	5.42	6.10	6.09	5.35	4.40	2.85	1.86	1.36	(44)	
(45)		RadStd	0.21	0.32	0.39	0.38	0.37	0.45	0.34	0.34	0.35	0.35	0.16	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(47) Regional (60 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(46) Station Only	(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page